PH7002: Concepts in Statistical Mechanics
Assignment 4 Solutions

Moments of momentum

We consider a gas of N classical particles of mass m in thermal equilibrium at a temperature T in $d = 3$.

(a) The equilibrium particle density $f(\mathbf{p}, \mathbf{q})$ for coordinate $\mathbf{q}$ and momentum $\mathbf{p}$ is given by

$$f(\mathbf{p}, \mathbf{q}) = \frac{n}{(2\pi m k_{\mathrm{B}} T)^{3/2}} \exp\left(-\frac{p^2}{2\pi m k_{\mathrm{B}} T}\right) .$$

With $n = 1$, it is straightforward to check that $\int_0^\infty dp\, f(p) = 1$.

(b) By evaluating $\int_0^\infty dp, p^2 f(p)$, we obtain $\langle p^2 \rangle = 3mk_{\mathrm{B}}T$. The average energy is therefore $\langle E \rangle = 3k_{\mathrm{B}}T/2$, which agrees with the equipartition theorem since we have three quadratic modes from the momentum.

(c) The joint characteristic function $\langle \exp(-i\mathbf{k} \cdot \mathbf{p}) \rangle$ can be obtained by evaluating the Gaussian average with Hubbard-Stratonovich transformation

$$\widetilde{p}(\mathbf{k}) = \langle \exp(-i\mathbf{k} \cdot \mathbf{p}) \rangle = \exp\left(-\frac{mk_{\mathrm{B}}T}{2} k^2\right) .$$

(d) The cumulants are calculated from the characteristic function as follow

$$\langle p_x^l p_y^m p_z^n \rangle = \left[\frac{\partial}{\partial(-ik_x)}\right]^l \left[\frac{\partial}{\partial(-ik_y)}\right]^m \left[\frac{\partial}{\partial(-ik_z)}\right]^n \log \widetilde{p}(\mathbf{k})\Big|_{\mathbf{k}=0} = mk_{\mathrm{B}}T\left(\delta_{l2}\delta_{m0}\delta_{n0} + \delta_{l0}\delta_{m2}\delta_{n0} + \delta_{l0}\delta_{m0}\delta_{n2}\right) .$$

So, the second cumulants are the only non-vanishing cumulants.

(e) We can calculate the joint moment using Wick's theorem:

$$\langle p_\alpha p_\beta (\mathbf{p} \cdot \mathbf{p}) \rangle = \langle p_\alpha p_\beta p_\gamma p_\gamma \rangle = \langle p_\alpha p_\beta \rangle \langle p_\gamma p_\gamma \rangle + 2\langle p_\alpha p_\gamma \rangle \langle p_\beta p_\gamma \rangle = (mk_{\mathrm{B}}T)^2 \left(\delta_{\alpha\beta}\delta_{\gamma\gamma} + 2\delta_{\alpha\gamma}\delta_{\beta\gamma}\right) = 5(mk_{\mathrm{B}}T)^2 \delta_{\alpha\beta}$$

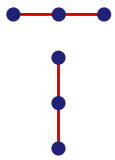
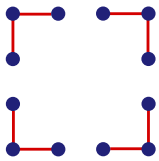
Yeomans 6.3

The exact result for the spontaneous magnetization per spin of the two-dimensional Ising model on a square lattice at low temperatures is

$$\langle s \rangle = 1 - 2u^2 - 8u^3 - 34u^4 - 152u^5 - 714u^6 - 3472u^7 - \ldots,$$

where $u = e^{-4\beta J}$.

(a) At low temperatures, almost all the spins are oriented in the same direction; low energy excitations correspond to islands of flipped spins. With non-zero external field H, the general behavior of the terms in a low-temperature expansion of the partition function is then of the form $e^{-2\beta J} e^{-2\beta H}$. We evaluate the series expansion based on enumeration of low-energy excitations above the ground state. Letting $u = e^{-4\beta J}$ and $v = e^{-2\beta H}$, we have

Configuration	Count	Boltzmann weight
	N	u^2v
	$2N$	u^3v^2
	$\frac{N}{2}(N-5)$	u^4v^2
	$2N$	u^4v^3
	$4N$	u^4v^3
	N	u^4v^4

The free energy per site for small u is then given by

$$\mathcal{F} \approx -(2J+H) - \frac{1}{\beta}\left(u^2v + u^3v^2 + u^4\left(-\frac{5}{2}v^2 + 6v^3 + v^4\right)\right),$$

and therefore the magnetization per site is

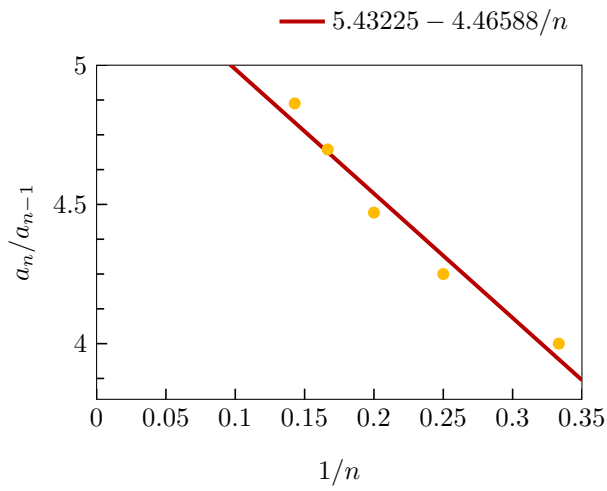
$$\langle s\rangle = -\frac{\partial \mathcal{F}}{\partial H} = 2\beta v\frac{\partial \mathcal{F}}{\partial v} = 1 - 2u^2 - 8u^3v - 2u^4\left(-5v + 18v^2 + 4v^3\right) .$$

In the absence of external field H, i.e. $v = 1$, we recover the exact solution.

(b) We can try to extract the parameters u_c and β_c from the calculated coefficients in the series by requiring that the ratio of successive terms to be

$$\frac{a_n}{a_{n-1}} = u_c^{-1}\left(1 + \frac{\beta - 1}{n}\right) .$$

Thus a plot of a_n/a_{n-1} versus $1/n$ should be a straight line with intercept u_c^{-1} and slope $u_c^{-1}(\beta - 1)$ (see figure below). From the fitting curve, we find $u_c \approx 0.184$ and $\beta \approx 0.178$ which are close to their exact values.



Yeomans 6.4

Note that the ground state is ferromagnetic with $s_i = 1$ for all spins. The expansion for the partition function is obtained by including low energy excitations around this state, which is a single overturned spin. Any sites can be chosen for the excitation, which corresponds to energy cost $2\beta J \cdot (2 \cdot 3)$ (with respect to the ground state). The next lowest energy excitation corresponds to a dimer of negative spins with energy cost of $2\beta J \cdot (4 \cdot 3 - 2)$ with multiplicity of $3N$ (there are 3 possible orientations for the dimer). The expansion reads

$$\mathcal{Z} = 2e^{3\beta NJ}\left(1 + Ne^{-12\beta J} + 3Ne^{-20\beta J} + \frac{N}{2}(N-7)e^{-24\beta J} + \cdots\right) .$$

The zeroth order term comes from the ground state, while the fourth term comes from flipping two disjoint single spins. The next higher order terms correspond to 3 spins connected ($e^{-28\beta J}$), 2 spins connected and 1 spin disjointed ($e^{-32\beta J}$), and 4 spins in a square form ($e^{-32\beta J}$). Below is the list of graphs, associated counts, and Boltzmann weights in terms of $x = e^{-2\beta J}$:

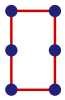
Configuration	Count	Boltzmann weight
	N	x^6
	$3N$	x^{10}
	$\frac{N}{2}(N-7)$	x^{12}
	$3N$	x^{14}
	$12N$	x^{14}
	$3N\,(N-12)$	x^{16}
	$3N$	x^{16}

High-temperature expansion of three-dimensional Ising model

(a) The partition function of three-dimensional Ising model on a simple cubic lattice is given by

$$Z = \sum_{\{s_i\}} e^{K \sum_{\langle ij \rangle} s_i s_j} = (\cosh K)^{3N} \sum_{\{s_i\}} \prod_{\langle ij \rangle} (1 + s_i s_j u) = 2^N (\cosh K)^{3N} \left(1 + a_4 u^4 + a_6 u^6 + \dots\right) \ .$$

Here, we've used the fact that the only non-vanishing terms are those that correspond to even powers of $u = \tanh K$. So, the graphs would be loops formed by $2, 4$, or 6 lines.

Configuration	Count	Boltzmann weight
	$3N$	u^4
	$6N$	u^6
	$12N$	u^6

(b) The free energy per site is given by

$$\beta \mathcal{F} = -\frac{1}{N} \log Z \approx -\log 2 - 3 \log \cosh K - 3 (\tanh K)^4 - 18 (\tanh K)^6 \ .$$